CRITICAL OUTAGE // SEV-1

INCIDENT ID: NX-2026-9941 | **FACILITY:** Melbourne Hyperscale Core Exchange | **TARGET:** x86_64 Linux / C++20

PACKET RUSH: EVALUATION MATRIX & RULES

Part 2: Competition Rules, Benchmark Contract & Scoring Model

1. Multi-Dimensional 100-Point Scoring System

Submissions are evaluated across four performance dimensions to ensure balanced, production-ready systems engineering:

Metric Dimension	Max Points	Production Target	Mathematical Scoring Formula
`Throughput (T)`	40 pts	25.0 Mpps	`S_T = 40 × min(1.0, T / 25.0)`
`P99 Batch Latency (L)`	25 pts	2.5 µs (64 pkts)	`S_L = 25 × min(1.0, 2.5 / L)`
`Peak Memory RSS (M)`	20 pts	32.0 MiB	`S_M = 20 × min(1.0, 32.0 / M)`
`Dynamic Updates (U)`	15 pts	5.0 µs	`S_U = 15 × min(1.0, 5.0 / U)`

ZERO-TOLERANCE CORRECTNESS MANDATE:
A fast classifier that returns incorrect actions is unacceptable in critical infrastructure. Any classification mismatch against the golden baseline, runtime crash (SIGSEGV/SIGABRT), or timeout awards **0.0 points** for that scenario and disqualifies the entry from the leaderboard.

2. Official Benchmark Contract

• **Execution Topology**: Single isolated x86_64 physical CPU core (fixed clock, no thermal throttling).

• **Concurrency Contract**: Single-threaded synchronous execution. Helper threads and background thread pools in classification are prohibited. Intra-thread SIMD (AVX2/AVX-512) is encouraged.

• **Throughput vs P99 Latency Contract**: At 25.0 Mpps, a 64-packet batch has an amortized budget of 2.56 µs. The P99 batch latency target (≤ 2.5 µs) enforces a strict tail-latency SLA under the exact same benchmark workload stream.

• **Memory Footprint**: Peak process RSS via ru_maxrss (≤ 32.0 MiB ceiling, including glibc arenas).

3. Hidden Test Scenarios

Scenario	Weight	Scale	Evaluation Objective
`01_correctness_small`	15%	1k rules / 5k pkts	Boundary checks, /0 default masks, ICMP Type 0 discrimination
`02_scalability_medium`	20%	15k rules / 20k pkts	Cache hierarchy pressure, testing sublinear lookup scaling
`03_performance_large`	25%	40k rules / 30k pkts	Throughput stress testing, vectorization & SIMD utilization
`04_adversarial_skew`	20%	20k rules / 20k pkts	Dense wildcards (65%+), nested CIDR, overlapping ports
`05_dynamic_heavy`	20%	15k rules + 10k ops	10,000 interleaved dynamic operations (5k inserts / 5k deletes) measuring update latency

4. Technical Regulations & Submission Rules

• **Compiler Environment**: g++ -O3 -std=c++20 -march=native -Wall -Wextra on Ubuntu 24.04 LTS x86_64.

• **Submission Artifacts**: Modified starter/classifier.h and starter/classifier.cpp.